Carlo Buccisano

Curriculum Vitae

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carlobuccisano.github.io
github.com/carlobuccisano
Italian, English, French and (un poco de) Spanish

PhD in pure mathematics looking for Machine Learning and Data Science industry roles.

Work experience

2024–2025 **Research and Teaching Assistant (ATER)**, *Université Paris-Saclay (Orsay)*
Teaching 192 hours of undergraduate courses in French: analysis, linear algebra and geometry.
Soft skills: communication and leadership.

2021–2025 **Ph.D. in Mathematics**, *Université Paris-Saclay (Orsay)*
I led research in derived algebraic geometry, a subject mixing algebraic geometry with modern homotopy theory techniques. More specifically, I studied derived D-modules and used them to connect several existing concepts in the literature. My thesis can be found on HAL and on arXiv.
Soft skills:
- I co-organized the GAeL XXIX conference, that brought more than 100 people to Orsay;
- I gave several public research talks and poster presentations to audiences of different backgrounds, strenghtening my communication skills.

July 2015 **Computer Programmer**, *IBM*, UK
- I worked paired with another colleague, part of a team of 5;
- I developped and documented some APIs of an internal planner tool in Python;
- this internship was awarded by Banca d'Italia (prize for the study of Maths and Computer Science).

Computer skills

- I coded my personal webpage using Hugo, for which I learnt HTML, CSS and Javascript.
- I can code in `C++` and `Python` (from competitive programming and undergraduate courses).
- I learnt and mastered LaTeX during my PhD, to write my thesis.
- I use Arch Linux as a daily driver: I am well versed in Bash scripting and Docker containers.
- I use `git` version control, both from the terminal and from GitHub web interface.

Education

2016–2022 **Galileian Diploma**, *Scuola Galileiana di Studi Superiori, Padova (Italy)*
Graduated from Scuola Galileiana, an honours college in Padova (7% admission rate).
Relevant courses: Machine Learning, Concurrent Java programming and game theory.

2019–2021 **M.Sc. in Mathematics**, *Università degli Studi di Padova & Université Paris-Saclay*
Graduated with maximum honours from Algant double degree, first year in Padova (30/30) then in Orsay (18.7/20). I was awarded a 10.000€ scholarship from FMJH.

2016–2019 **B.Sc. in Mathematics**, *Università degli Studi di Padova*
GPA: 29.965/30; *110/110 cum laude* (maximum honours).

Awards

2013–2015 **Medals in Italian Maths and Informatics Olympiads**
I obtained two gold (an absolute third and first place) and one bronze medals in Italian Informatics Olympiads, and one silver medal in Italian Mathematical Olympiads.
Skills: problem solving, work under pressure, data structures and algorithms.

2013–2015 **Competitive programming**
Participated in online competitions such as Google Code Jam, HashCode, Codeforces, USACO.